

Face Attendance AI

Production-Ready AI Face Recognition Attendance System — Project Documentation

1. Executive Summary & Problem Statement

Traditional attendance methods like paper registers, RFID cards, or fingerprint scanners suffer from critical drawbacks: they are time-consuming, prone to 'buddy punching' (proxy attendance), and unhygienic in post-pandemic work environments. **Face Attendance AI** is an automated, contactless attendance solution combining Computer Vision, Deep Learning, and a high-performance Web application to identify faces in real-time, log timestamps, prevent duplicate entries, and visualize analytics.

2. Complete Technology Stack

Component	Technology / Library	Key Responsibility & Justification
Backend Server	Python 3.14 + FastAPI + Uvicorn	High-performance asynchronous REST API, WebSockets handler, static file server, and background processing.
Computer Vision & AI	OpenCV + Face Recognition	Extracts 128-dimensional deep facial landmarks, calculates Euclidean metric distance, and verifies identities.
Web Frontend (SPA)	HTML5 + Vanilla CSS3 + JavaScript	Cyber-luxe glassmorphic dashboard, responsive layout, theme toggle (Dark/Light), zero bulky frontend build dependencies.
Live Video & HUD	HTML5 WebRTC + Canvas API	Browser-level 60 FPS live video stream, dynamic corner reticles, laser scanning overlay, and bounding boxes.
Real-Time Live Sync	WebSockets (/ws/live)	Instant multi-client live updates without page refresh upon every attendance event.
Audio & Speech Engine	Web Audio API + Web Speech API	Synthesizes futuristic check-in audio chimes and speaks automated welcome greetings aloud to employees.
Persistence Layer	Attendance.csv + JSON Storage	Direct two-way compatibility with Excel/payroll CSV files, enriched with photo snapshots and role metadata.

3. System Architecture & Workflow

- Step 1: Face Enrollment:** Face images are uploaded via the dashboard or placed in the Images/ directory. The system generates a unique 128-dimensional embedding vector for each person.
- Step 2: Live Detection:** When a person approaches the camera, WebRTC captures video frames at 60 FPS. The browser HUD locks a tracking reticle over their face.
- Step 3: Matching & Verification:** The live face vector is compared against known roster vectors using Euclidean distance. If distance < 0.55 (confidence > 95%), a match is confirmed.
- Step 4: Anti-Duplicate Cooldown:** The backend validates an in-memory timer (e.g. 300s). Redundant check-ins within the cooldown period are gracefully acknowledged without logging duplicate lines.

Step 5: Status Tagging (On-Time vs Late): The check-in time is compared against the configured Official Start Time (e.g. 09:00 AM) + Grace Period (15m). Arrivals before 09:15 AM receive **On Time** (green); arrivals after receive **Late** (red).

Step 6: Real-Time Sync & Audio Feedback: The event is appended to Attendance.csv, broadcasted via WebSockets to all dashboards, and an audible greeting speaks: *'Welcome [Name]! Attendance registered.'*

4. Core Dashboard Modules

- **AI Scanner Kiosk:** Live webcam viewport with cyber laser HUD, device selector, FPS meter, and live feed.
- **Attendance Audit Table:** Searchable, filterable by date/department/status, with instant CSV/Excel export.
- **Staff Face Directory:** Visual employee cards with photo avatars, last seen timestamps, and '+ Enroll' modal.
- **Analytics Hub:** Real-time KPI tiles (Total Enrolled, Present Today, Attendance %, On-Time %) and 24H traffic charts.
- **Settings Panel:** Configurable start times, grace period, anti-duplicate cooldown slider, and Dark/Light theme.

5. Step-by-Step Usage & Presentation Guide

- 1. Launch Server:** Double-click run_server.bat in the project directory. The browser opens automatically to <http://localhost:8000>.
- 2. Enroll Colleague:** Go to 'Staff Roster' → Click '+ Enroll New Person' → Take a webcam photo or upload an image file → Click 'Save & Enroll'.
- 3. Mark Attendance:** Go to 'AI Scanner Kiosk' → Stand in front of camera → System detects face, plays sound chime, and marks attendance.
- 4. Download Report:** Go to 'Attendance Logs' → Click 'Export CSV' to download an Excel-ready attendance sheet.

6. Key Viva & Exam Questions (Ready Answers)

Question	Recommended Answer
How does the AI recognize faces?	It extracts 128 deep facial landmark measurements (embeddings) and calculates Euclidean distance to find the closest matching enrolled identity.
How are duplicate check-ins prevented?	The backend enforces a configurable cooldown throttle cache (e.g. 5 minutes). Duplicate scans within this window are rejected from re-logging.
How is Late vs On-Time determined?	The check-in timestamp is compared against the Official Start Time plus Grace Period configured in the system rules.
Why use WebSockets?	WebSockets enable bidirectional real-time event broadcasting so multiple kiosks and admin screens update instantly without polling or page refresh.